

Aadarsh Gaikwad

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Education

Northeastern University

Boston, MA

MS in Data Science

Jan 2024–May 2025

Experience

J.P. Morgan Chase & Co

Data Scientist

Apr 2023 – Dec 2023

- Improved loan document classification accuracy from 78% to 86% by migrating existing sklearn models to **BERT** architecture using **PyTorch** and **Hugging Face Transformers**.
- Addressed class imbalance across 8 document types by implementing **SMOTE**, focal loss, and stratified sampling on 50K labeled documents, achieving consistent F1 scores above 80% for minority classes.
- Extended pipeline capabilities with **spaCy NER** to extract loan amount, applicant details, and property information, reducing manual data entry time by 40%.
- Contributed to document processing API using **Node.js Lambda** and **API Gateway** to handle secure uploads and trigger batch inference with results in **DynamoDB**.

Data Engineer

Mar 2022 – Apr 2023

- Built attribution data pipelines processing 3+ TB of cross-channel customer data using **AWS Step Functions** to orchestrate **EMR, Lambda** and **Glue** jobs.
- Created pipeline control APIs with **Node.js** and Lambda exposed through **API Gateway**, enabling Step Functions triggering and job status monitoring.
- Optimized **PySpark** jobs and EMR tuning with **Spot Instances**, reducing compute costs by \$50K annually.
- Designed **Snowflake** architecture using Snowpipe, clustering keys, and materialized views for real-time attribution reporting.
- Automated deployments using **Docker, Terraform**, and **GitHub Actions** with blue-green strategy, reducing release times.

Wells Fargo

Data Engineer

Apr 2021 – Mar 2022

- Supported fraud detection initiatives by building **SageMaker** data pipelines that delivered curated feature datasets to ML teams.
- Implemented feature engineering using **PySpark** within **SageMaker Processing** including one-hot encoding, scaling, and time-based aggregations.
- Created data quality validation framework using **Great Expectations**, checking for nulls, outliers, and schema consistency.
- Configured **S3 Intelligent-Tiering** to move infrequently accessed data between storage tiers, reducing storage costs by 20%.
- Implemented **CloudWatch** logging and **SNS** alerts to ensure pipeline SLA compliance and traceability.

Cognizant Technology Solutions

Data Engineer

Jul 2018 – Apr 2021

- Engineered regulatory reporting pipelines processing 1 TB+/month of financial and compliance data with **PySpark** and **Hive**.
- Designed **REST API** microservices with **FastAPI** and **SQLAlchemy** to migrate data from **OLTP** systems to **Hadoop**.
- Enforced **GDPR** using **tokenization** and **column-level encryption**, aligning with enterprise-wide data governance policies.
- Secured internal APIs using **OAuth 2.0** and **JWT**, added **rate limiting**, and conducted testing using **Postman** and **Pytest**.
- Improved **PySpark SQL** query performance using **broadcast joins**, **column pruning**, and **predicate pushdown** techniques.
- Delivered monthly data reports and **Tableau** dashboards to leadership, communicating attribution impact and business KPIs.

Certifications

Amazon Web Services: AWS Certified Data Engineer – Associate

[Link]

Open Source Contribution

prompt-to-json: Developed and packaged a library that converts free-form prompts into structured JSON using **OpenAI**; released on **PyPI** with **pre-commit** hooks, **GitHub Actions CI/CD**, and **code coverage** integration. [GitHub] [PyPI]

Projects

Research Assistant:

- Designed a retrieval-augmented generation (RAG) pipeline using **AWS Bedrock** (Llama-3-70B), **LangChain** and **ChromaDB** to generate context-aware responses from historical support data and knowledge base.
- Built web data pipelines with **BeautifulSoup** and **Scrapy**, storing structured outputs in **Amazon S3** after validation.

Skills

Languages: Python, SQL, Shell Scripting, JavaScript, Node.js

Data Platforms: Snowflake, Redshift, BigQuery, Hive, Spark/PySpark

Orchestration & MLOps: Airflow, dbt, Git, GitHub Actions, Terraform, Docker, EMR, Glue, Step Functions, SageMaker

Databases: MySQL, Oracle, PostgreSQL, MongoDB, DynamoDB, ChromaDB

AI: LLM fine-tuning (PEFT, QLoRA), Quantization, Prompt Engineering, RAG, Guardrails, Ollama, LangChain